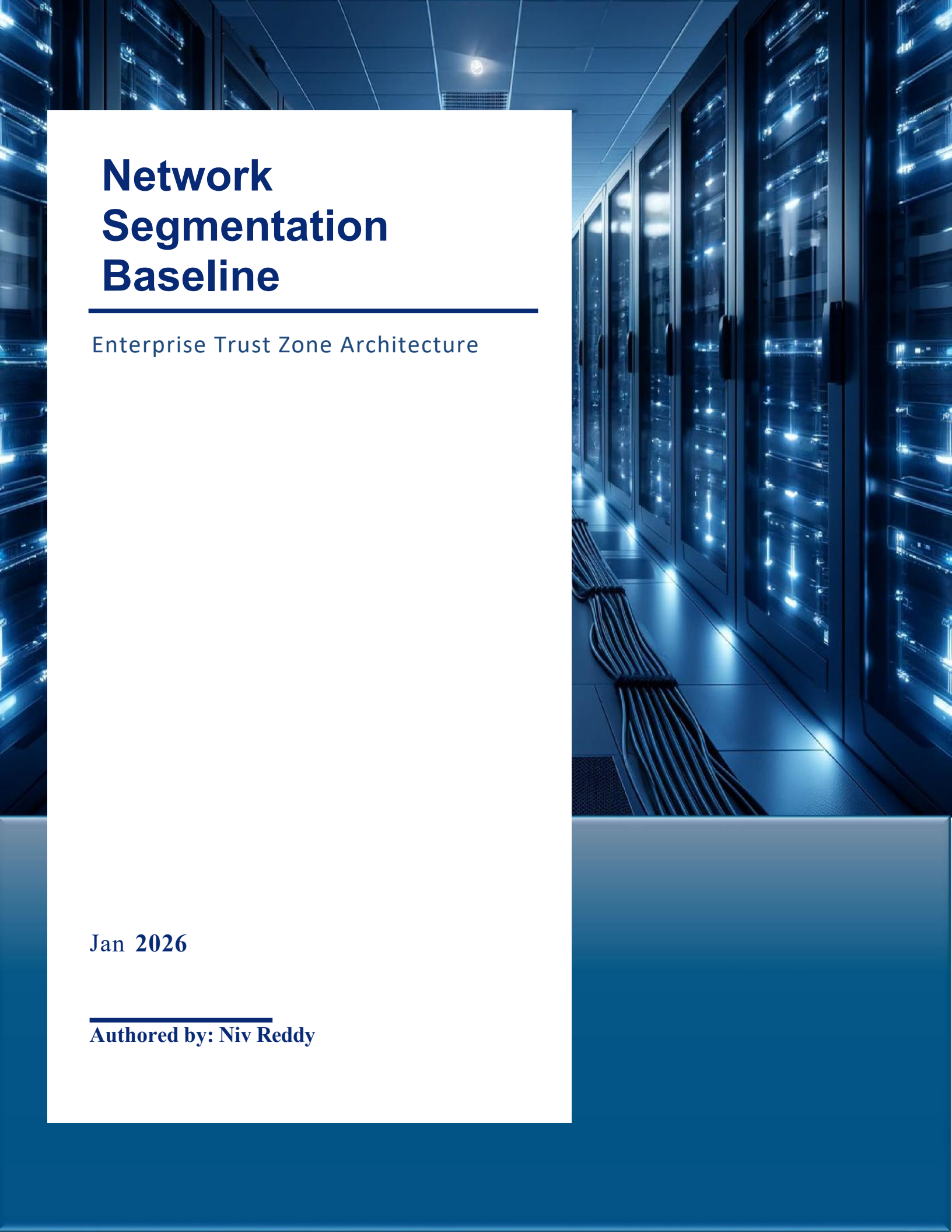




Network Segmentation Baseline

Enterprise Trust Zone Architecture

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Introduction

Context

This document defines the baseline network segmentation architecture of the lab.niv.local enterprise environment.

It records the first physical enforcement layer that establishes trust boundaries between enterprise systems, user endpoints, and future security control planes.

This segmentation layer forms the foundation for:

- Lateral movement prevention
- Privilege boundary enforcement
- Controlled routing and firewall policy
- Detection engineering and adversary simulation
- Zero Trust security modeling

All future firewalling, routing, detection, and attack simulation layers anchor to this segmentation fabric.

Enterprise Segmentation Fabric

Before any firewalling or routing was introduced, a dedicated Layer-2 enterprise backbone was engineered to provide physical enforcement of trust zones.

This backbone operates as a virtual enterprise switch that carries multiple isolated security domains while remaining sealed from the residential LAN and internet.

Fabric Implementation

A VLAN-aware Proxmox bridge was engineered to serve as the corporate backbone

```
auto vmbr1
iface vmbr1 inet manual
    bridge-ports none
    bridge-stp off
    bridge-fd 0
    bridge-vlan-aware yes
    bridge-vids 2-4094
```

Resulting Trust Boundary

The segmentation fabric provides:

- A sealed corporate broadcast domain
- VLAN-based Layer-2 isolation
- No routing or gateway exposure
- No residential DNS or DHCP exposure
- Deterministic physical enforcement of trust boundaries

All enterprise systems attach exclusively to this fabric.

Trust Zone Model

Trust Zone	VLAN	Purpose
Server Trust Zone	10	Identity infrastructure and core services
Workstation Trust Zone	20	User endpoints and client devices
Security/Admin Trust Zone	30	Security control plane and administrative tooling

At the conclusion of Phase 2, VLAN 10 and VLAN 20 are fully enforced

VLAN 30 is reserved for Phase 3 security perimeter deployment

Server Trust Zone Establishment

Purpose

This checkpoint establishes the Tier-0 Server Trust Zone which protects identity, authentication, and directory infrastructure from lateral movement.

Implementation

DC01 was classified into the Server Trust Zone by tagging its NIC with VLAN 10 on the enterprise backbone.

Identity Stability Validation

- Static IP preserved: 10.0.10.10
- DNS resolution remained authoritative
- Kerberos AES-256 ticketing validated
- DC01 remained the KDC and forest root authority

Isolation Validation

A disposable test VM was placed in the untagged network and manually addressed. It was unable to reach DC01

This validated physical Layer-2 isolation between VLAN 1 and VLAN 10.

Result

- Tier-0 identity infrastructure is physically isolated
- Lateral movement into identity services is blocked
- Server Trust Zone is live and enforced

Workstation Zone Establishment

Purpose

This checkpoint establishes the Workstation Trust Zone, which contains all user endpoints and represents the primary lateral-movement risk surface.

Implementation

A dedicated workstation VM, BLUE-WS01, was deployed and assigned VLAN 20

No routing, DHCP, or gateway services exist on this VLAN

Network State Validation

BLUE-WS01 was validated:

- Automatically assigned APIPA address (169.254.x.x)
- No default gateway present
- No DNS servers assigned
- No outbound routing exists

This confirms VLAN 20 is a sealed internal user network.

Isolation Validation

ping 10.0.10.10 = General failure

BLUE-WS01 cannot reach DC01.

This confirms physical isolation between the Workstation Trust Zone and the Server Trust Zone.

Result

- User endpoints are physically isolated
- No lateral movement into Tier-0 infrastructure is possible
- VLAN-based segmentation is live and enforced